RK356X System Suspend Configuration Guide

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Rockchip Electronics Co., Ltd.

Building 18, Area A, Software Park, Tongpan Road, Fuzhou, Fujian, P.R.C.

Website: www.rock-chips.com

Customer service Tel: +86-4007-700-590

Customer service Fax: +86-591-83951833

Customer service e-Mail: fae@rock-chips.com

Preface

Overview

This document guides users on configuring the RK356X system suspend mode according to product requirements.

Product Version

Chip Models	Kernel Version
RK3568	4.19, 5.10

Intended Audience

This document (this guide) is primarily intended for the following engineers:

Technical Support Engineers

Software Development Engineers

Revision History

Version	Author	Date	Modification Description
V1.0.0	Xu Shengfei	2022-01-08	Initial version

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1. System Suspend

For all SoC platforms equipped with Trust, system suspend operations are handled within the Trust firmware. Since Trust implementations for system suspend vary across different platforms, **suspend configuration options and methods are not correlated or referenceable between platforms; this document applies exclusively to the RK356X platform**.

The system suspend process typically involves the following operations: powering down power domains and module IPs, disabling clocks and PLLs, entering DDR Self-refresh, switching the system bus to a low-speed clock (24M or 32K), cutting off vdd_arm/vdd_log power, and configuring wake-up sources. To meet the suspend mode requirements of different products, relevant configurations are currently passed to Trust during the boot phase via DTS nodes.

1.1 Driver Files

```
./drivers/soc/rockchip/rockchip_pm_config.c
./drivers/firmware/rockchip_sip.c
./include/dt-bindings/suspend/rockchip-rk3568.h
```

1.2 DTS Node

```
rockchip_suspend: rockchip-suspend {
    compatible = "rockchip,pm-rk3568";
    status = "okay";
    // Suspend log switch configuration: 0 disables logging, 1 enables logging
    rockchip,sleep-debug-en = <1>;
    // Standard configuration
    rockchip,sleep-mode-config = <
        (0
        | RKPM_SLP_ARMOFF_LOGOFF
        | RKPM_SLP_CENTER_OFF
        | RKPM_SLP_HW_PLLS_OFF
        | RKPM_SLP_PMUALIVE_32K
        | RKPM_SLP_OSC_DIS
        | RKPM_SLP_PMIC_LP
        | RKPM_SLP_32K_PVTM
        )
    >;
    // Wake-up source configuration
    rockchip,wakeup-config = <
        (0
        | RKPM_GPIO_WKUP_EN
        )
```

```
        >;
};
```

2. DTS Configuration

Currently supported configuration options are defined in:

```
./include/dt-bindings/suspend/rockchip-rk3568.h
```

2.1 Standard Configuration

Configuration items:

```
rockchip,sleep-mode-config = <...>;
```

Configuration source:

```
// The CPU is in WFI state during suspend, used only for debugging
#define RKPM_SLP_WFI                    BIT(0)
// Suspend: cpu_pd power down
#define RKPM_SLP_ARMOFF                 BIT(1)
// During suspend, DDR enters self-refresh and remains in retention state
#define RKPM_SLP_CENTER_OFF             BIT(2)
// Suspend logic power-off
#define RKPM_SLP_ARMOFF_LOGOFF          BIT(3)
// Support U-Boot low-power mode
#define RKPM_SLP_FROM_UBOOT             BIT(4)
// Suspend: PMIC enters low-power mode
#define RKPM_SLP_PMIC_LP                BIT(5)
// Suspend PLL poweroff
#define RKPM_SLP_HW_PLLS_OFF            BIT(6)
// Switch PMU PD to 32k during suspend
#define RKPM_SLP_PMUALIVE_32K           BIT(7)
// Suspend OSC off
#define RKPM_SLP_OSC_DIS                BIT(8)
// The 32K clock source for suspend is provided externally
#define RKPM_SLP_32K_EXT                BIT(9)
// The 32k clock source for suspend is generated by the internal PVTM
#define RKPM_SLP_32K_PVTM               BIT(10)
```

2.2 Wake-up Source Configuration

Configuration items:

```
rockchip,wakeup-config = <...>;
```

Configuration source:

```
// support enable_irq_wake Interrupt wake-up
#define RKPM_CPU0_WKUP_EN               BIT(0)
// Support GPIO wake-up
#define RKPM_GPIO_WKUP_EN               BIT(4)
// Support UART0 wake-up, the 24M clock must remain enabled.
#define RKPM_UART0_WKUP_EN              BIT(5)
// Support SDMMC wake-up
#define RKPM_SDMMC0_WKUP_EN             BIT(6)
#define RKPM_SDMMC1_WKUP_EN             BIT(7)
#define RKPM_SDMMC2_WKUP_EN             BIT(8)
// Support USB wake-up
#define RKPM_USB_WKUP_EN                BIT(9)
// Support PCIe wake-up
#define RKPM_PCIE_WKUP_EN               BIT(10)
// Support VAD wake-up
#define RKPM_VAD_WKUP_EN                BIT(11)
// Support TIMER wake-up, the 24M clock must remain enabled.
#define RKPM_TIMER_WKUP_EN              BIT(12)
// Support PWM0 wake-up, the 24M clock must remain enabled.
#define RKPM_PWM0_WKUP_EN               BIT(13)
// Support TIMEOUT wake-up, typically used for debugging
#define RKPM_TIMEOUT_WKUP_EN            BIT(14)
// USB protocol wake-up support requires the 24M clock to remain enabled
#define RKPM_USB_LINESTATE_WKUP_EN      BIT(16)
```

Wake-up Source Considerations:

- RKPM_GPIO_WKUP_EN:

Only the GPIO0 pin group is supported as a wake-up source. For hardware design, it is recommended to assign required wake-up sources to the GPIO0 pin group.

2.3 Debug Configuration

Configuration Options:

```
rockchip,sleep-debug-en = <...>;
```

Debug considerations:

- Setting the value to 1 enables the debug function, which prints ATF suspend and resume logs during the suspend-resume cycle.

3. Log Output

The following section briefly explains the meaning of Trust log messages during system suspend and resume. For clarity, some log content is split across multiple lines in the examples below. Different suspend power modes may also produce different log outputs; actual displayed content shall prevail.

RK356X System Suspend Logs:

```
// abcdegh is a flag indicating the current step in the suspend process
// sleep mode config
INFO:           sleep mode config[0x5ec]:
INFO:           mode: RKPM_SLP_CENTER_PD
INFO:           mode: RKPM_SLP_ARMOFF_LOGOFF
INFO:           mode: RKPM_SLP_PMIC_LP
INFO:           mode: RKPM_SLP_HW_PLLS_PD
INFO:           mode: RKPM_SLP_PMUALIVE_32K
INFO:           mode: RKPM_SLP_OSC_DIS
INFO:           mode: RKPM_SLP_32K_PVTM
// Supported wake-up sources
INFO:    wakeup source config[0x10]:
INFO:           Enable GPIO0 interrupt as wakeup source
```

RK356X System Wake-up Log:

```
// Wake-up print
INFO:    [TEST]: wakeup count: 0
INFO:    wake interrupt ID: 65
INFO:    WAKEUP SOURCE: 0x10
INFO:           GPIO0 interrupt as wakeup source
INFO:                   GPIO0A3
// Wake-up source
GPIO interrupt wakeup
gpio0a3 Wake up the system via Interrupt
```